

Long-Term Efficacy and Safety of Fremanezumab in Migraine: Results of a 1-Year Study

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Disclosures

- **Peter J. Goadsby:** Reports grants and personal fees from Amgen and Eli Lilly and Company, and personal fees from Alder Biopharmaceuticals, Allergan, Autonomic Technologies Inc., Biohaven Pharmaceuticals Inc., Dr Reddy's Laboratories, Electrocore LLC, eNeura, Impel NeuroPharma, MundiPharma, Novartis, Teva Pharmaceuticals, Trigemina Inc., WL Gore, medicolegal work, Massachusetts Medical Society, UpToDate, Oxford University Press, and Wolters Kluwer; he holds a patent for magnetic stimulation for headache assigned to eNeura without fee.
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- **Paul P. Yeung:** Former employee of Teva Pharmaceuticals.
- **Joshua M. Cohen:** Employee of Teva Pharmaceuticals.
- **Ronghua Yang:** Employee of Teva Pharmaceuticals.

Background

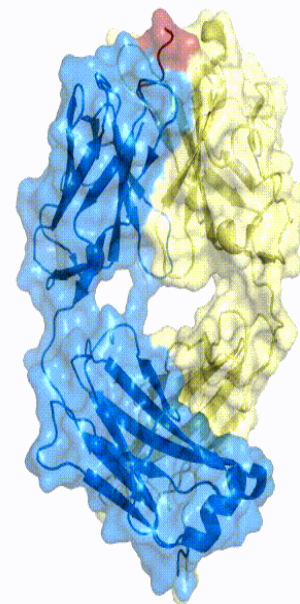
Benefits of a Long-Term Trial

- Longer treatment periods provide more-**stable estimates** of outcome measures¹
- **Full therapeutic potential** may only be realized over a longer duration of treatment, as efficacy for many medications accrues gradually¹
- Observations over long-term treatment periods may identify **additional adverse events** (AEs), and useful information about **treatment discontinuation**¹
- For medications targeting **chronic diseases** such as migraine², assessing the long-term benefits of treatment is especially important

Background and Objective

- **Fremanezumab**, a fully humanized monoclonal antibody (IgG2Δa) that selectively targets calcitonin gene-related peptide (CGRP), is approved in the US and the EU for the preventive treatment of migraine in adults^{1–4}
 - The Phase 3 HALO trials evaluated fremanezumab for the preventive treatment of chronic migraine (CM) and episodic migraine (EM)^{5,6}
 - Patients from the initial HALO CM and EM trials had the option of continuing treatment in the long-term HALO study (“rolling over”), and additional patients were allowed to directly enroll in the long-term study

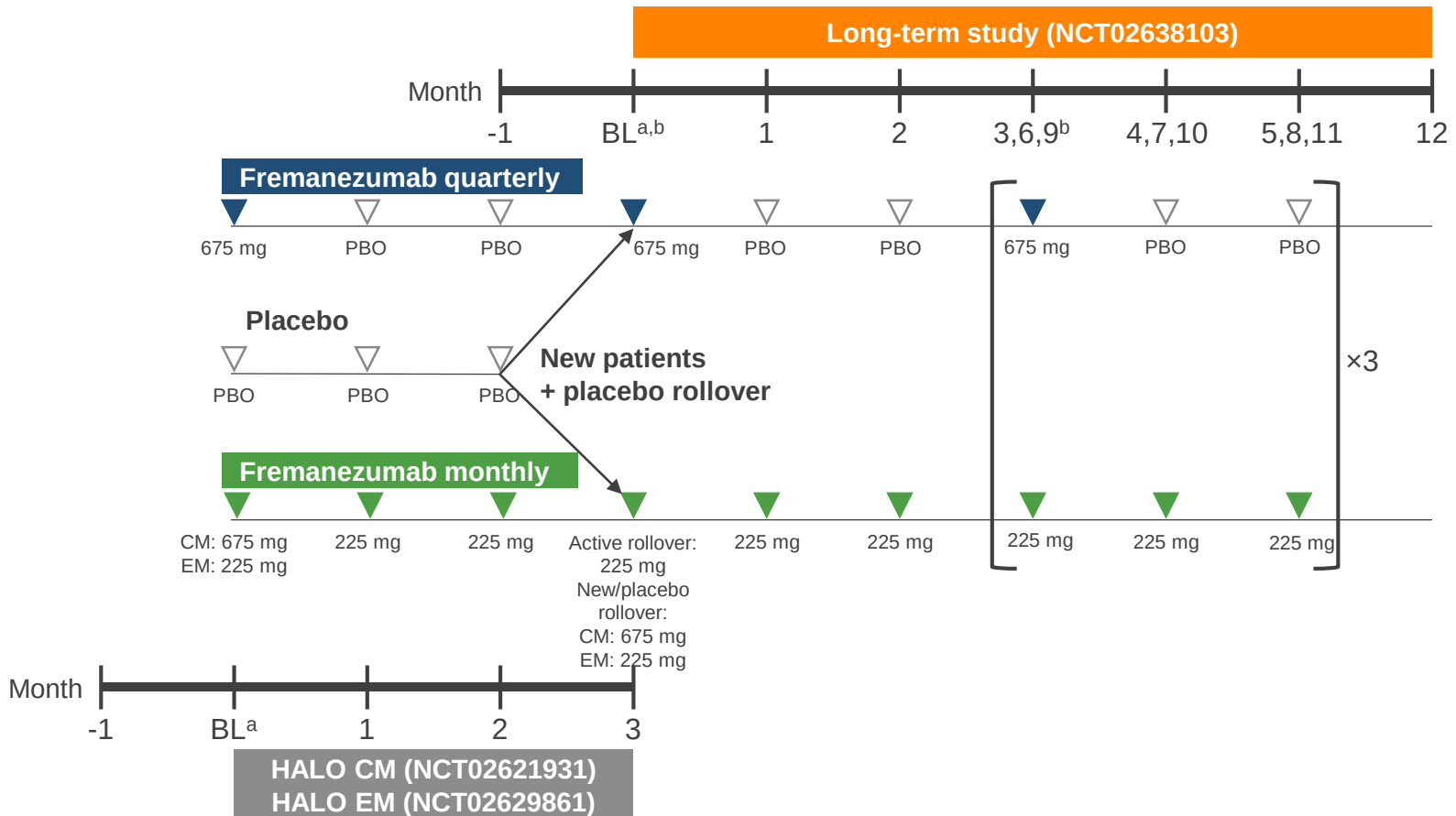
Fremanezumab



Objective: To evaluate the long-term efficacy and safety of fremanezumab in adults with migraine during the 12-month Phase 3 study

Long-Term HALO Trial: Study Design

- 12-month, multicenter, randomized, double-blind, parallel-group Phase 3 study
 - Patients who rolled over from the Phase 3 HALO CM and EM trials
 - Additional subset of new patients who were not previously enrolled in the HALO trials



^aBaseline refers to the 28-day pre-treatment period/Day 0 for the double-blind placebo-controlled study for rollover patients, and the 28-day pre-treatment period/Day 0 of the long-term study for new patients. ^bTo maintain the study blind, all patients received three injections at BL and Months 3, 6, and 9; patients received one injection at all other visits. BL, baseline; CM, chronic migraine; EM, episodic migraine; PBO, placebo.

Key Study Eligibility Criteria

Inclusion Criteria

- 18–70 years of age
- For CM:
 - Headache on ≥ 15 days per month
 - ≥ 8 days fulfilling International Classification of Headache Disorders [ICHD]-3 beta criteria for migraine, probable migraine, or use of triptan or ergot medications
- For EM:
 - Headache on 6–14 days (rollover patients) or 4–14 days (new patients)
 - ≥ 4 days fulfilling ICHD-3 beta criteria for migraine, probable migraine, or use of triptan or ergot medications
- Maximum of one (rollover patients) or two (new patients) concomitant migraine preventive medications allowed during the study
 - Must have at least moderate efficacy
 - Dosing must have been stable for ≥ 2 consecutive months prior to screening

Exclusion Criteria

- Only for patients rolling over from the previous HALO trials:
 - Use of onabotulinumtoxinA in the 4 months before screening
 - Use of opioids or barbiturates on >4 days during the pre-treatment period
 - Use of interventions or devices for migraine in the 2 months before screening
 - Previous failure in ≥ 2 of four of the following medication clusters after ≥ 3 months of treatment for CM or EM: divalproex sodium and sodium valproate; flunarizine and pizotifen; amitriptyline, nortriptyline, venlafaxine, and duloxetine; or atenolol, nadolol, metoprolol, propranolol, and timolol

Long-Term HALO Trial: Endpoints and Statistical Analysis

Efficacy Endpoints



- Mean change from baseline (28-day pre-treatment period) in the monthly average number of:
 - Migraine days
 - Headache days of at least moderate severity
 - Headache days of any severity
- Proportion of patients with a $\geq 50\%$ reduction in the monthly number of migraine days

Safety and Tolerability Endpoints



- Adverse events (AEs) and systematic local injection-site assessments (immediately and at 1 hour post-injection)

Statistical Analysis



- Efficacy analyses were conducted in the full analysis set (FAS): randomized patients who received ≥ 1 dose of study drug and had ≥ 1 post-baseline efficacy assessment
- The safety population included randomized patients who received ≥ 1 dose of study drug during the study

Baseline Demographics and Disease Characteristics^a

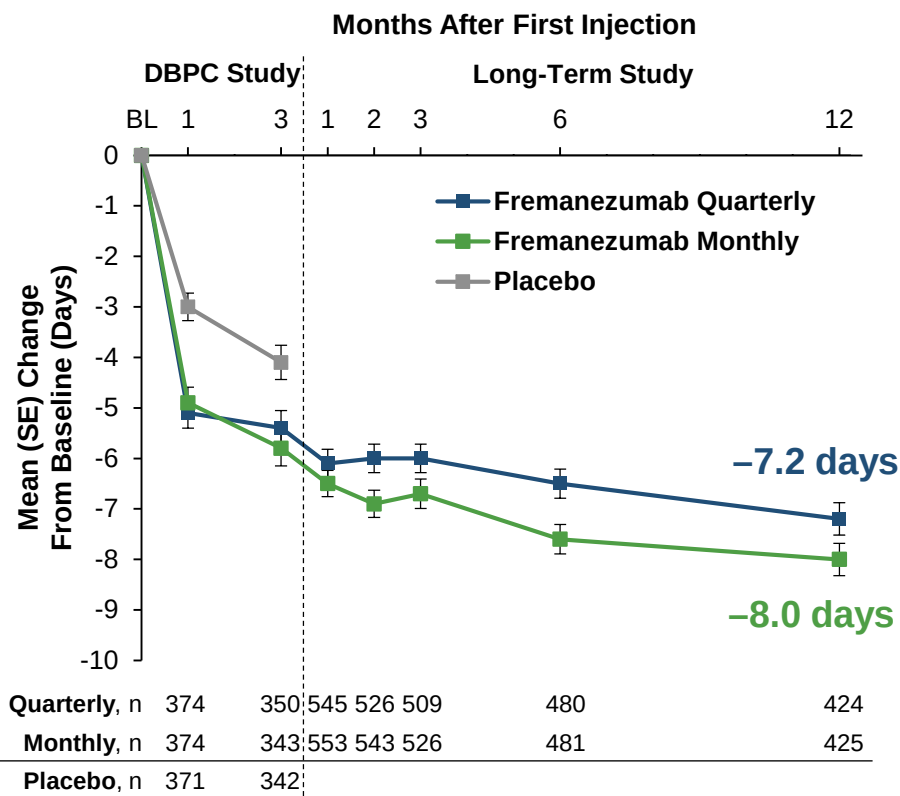
A total of 1890 patients were randomized to or rolled over into the study (rollover patients, n=1578; new patients, n=312), and 79% completed treatment

	CM		EM	
	Fremanezumab Quarterly (n=551)	Fremanezumab Monthly (n=559)	Fremanezumab Quarterly (n=394)	Fremanezumab Monthly (n=386)
Age, years, mean (SD)	43.7 (12.0)	42.6 (11.8)	43.3 (11.3)	44.7 (12.2)
Sex, female, n (%)	484 (88)	494 (88)	342 (87)	325 (84)
Years since initial migraine diagnosis, mean (SD)	21.7 (13.4)	21.2 (12.1)	21.6 (12.7)	21.9 (12.8)
Current preventive medication use, n (%)	128 (23)	142 (25)	89 (23)	92 (24)
Current acute headache medication use, n (%)	528 (96)	528 (94)	382 (97)	369 (96)
Prior topiramate use, n (%)	205 (37)	204 (36)	88 (22)	100 (26)
Prior onabotulinumtoxinA use, n (%)	127 (23)	122 (22)	28 (7)	27 (7)
Migraine days, mean (SD) ^b	16.4 (5.1)	16.4 (5.3)	9.2 (2.6)	9.1 (2.7)
Headache days of at least moderate severity, mean (SD) ^c	13.7 (5.6)	13.5 (6.0)	7.4 (3.2)	7.3 (3.0)
Headache days of any severity, mean (SD) ^d	16.2 (5.8)	16.3 (6.2)	8.6 (3.2)	8.5 (3.1)
Headache days of any severity and duration, mean (SD)	20.4 (4.1)	20.5 (4.4)	11.3 (2.4)	10.9 (2.6)

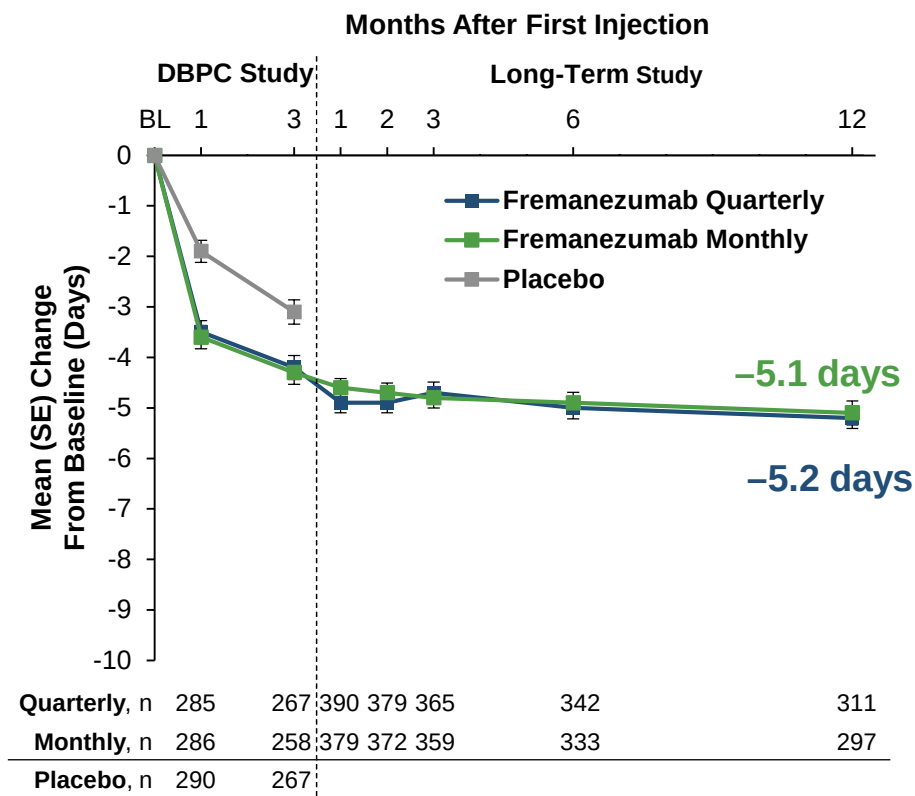
^aIn the ITT population, defined as all randomized/rolled-over patients. ^bA calendar day in which the patient reported either headache pain that lasted ≥4 hours (CM) or ≥2 hours (EM) consecutively, which met criteria for migraine or probable migraine; or a day when a headache of any duration was treated with migraine-specific medications (triptans or ergots). ^cA calendar day in which the patient reported either a day with headache pain that lasted ≥4 hours consecutively with a peak severity of at least moderate severity, or a day when acute migraine-specific medications (triptans or ergots) were used to treat a headache of any severity or duration. ^dA calendar day in which the patient reported headache pain of any severity that lasted ≥4 hours consecutively. CM, chronic migraine; EM, episodic migraine; ITT, intent-to-treat; SD, standard deviation.

Long-Term Change in Monthly Average Number of Migraine Days in Patients With CM and EM^a

Chronic Migraine



Episodic Migraine

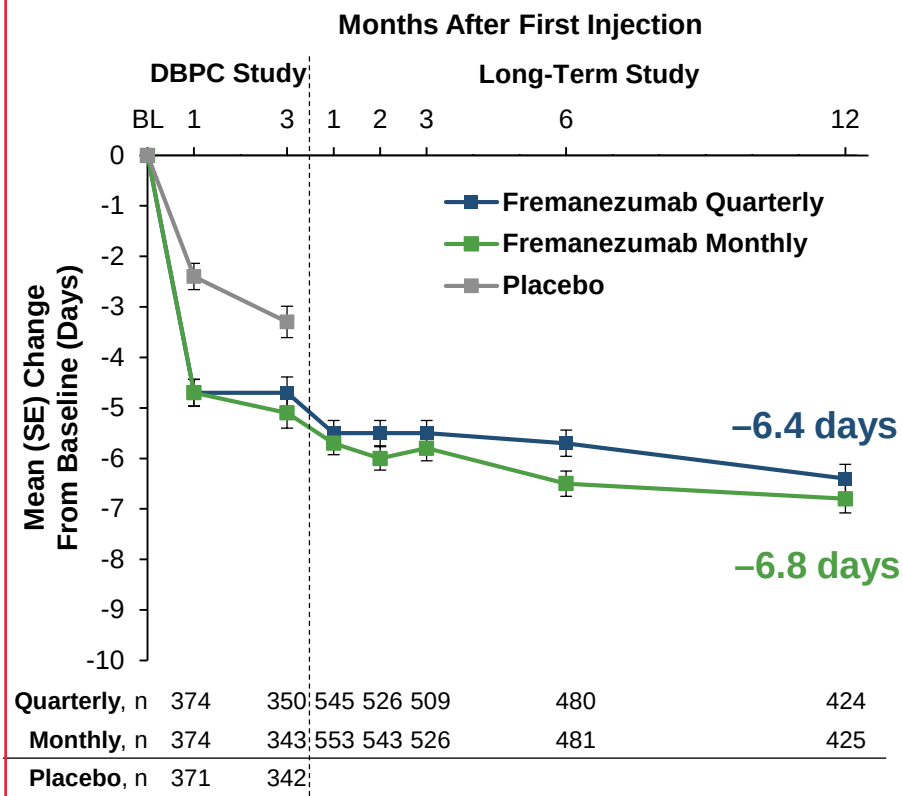


Treatment with fremanezumab quarterly or monthly resulted in sustained reductions from baseline in the **monthly average number of migraine days**

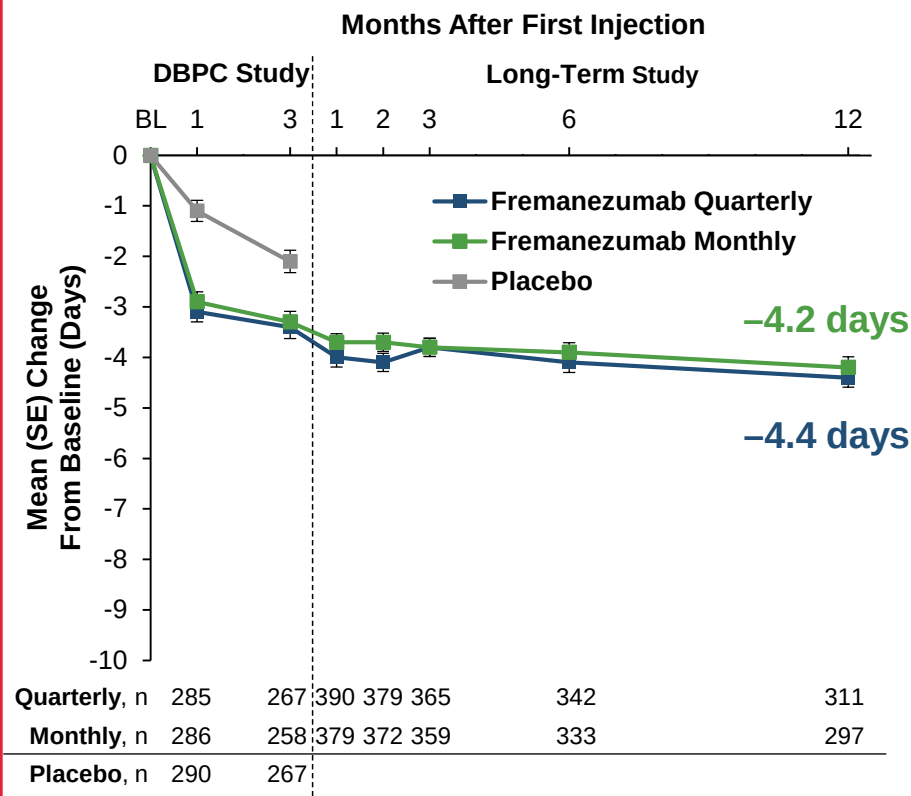
^aIn the FAS population, defined as randomized subjects who received ≥ 1 dose of study drug and had ≥ 1 post-baseline efficacy assessment. BL, baseline; CM, chronic migraine; DBPC, double-blind, placebo-controlled; EM, episodic migraine; FAS, full analysis set; SE, standard error.

Long-Term Change in Monthly Average Number of Headache Days of at Least Moderate Severity in Patients With CM and EM^a

Chronic Migraine



Episodic Migraine

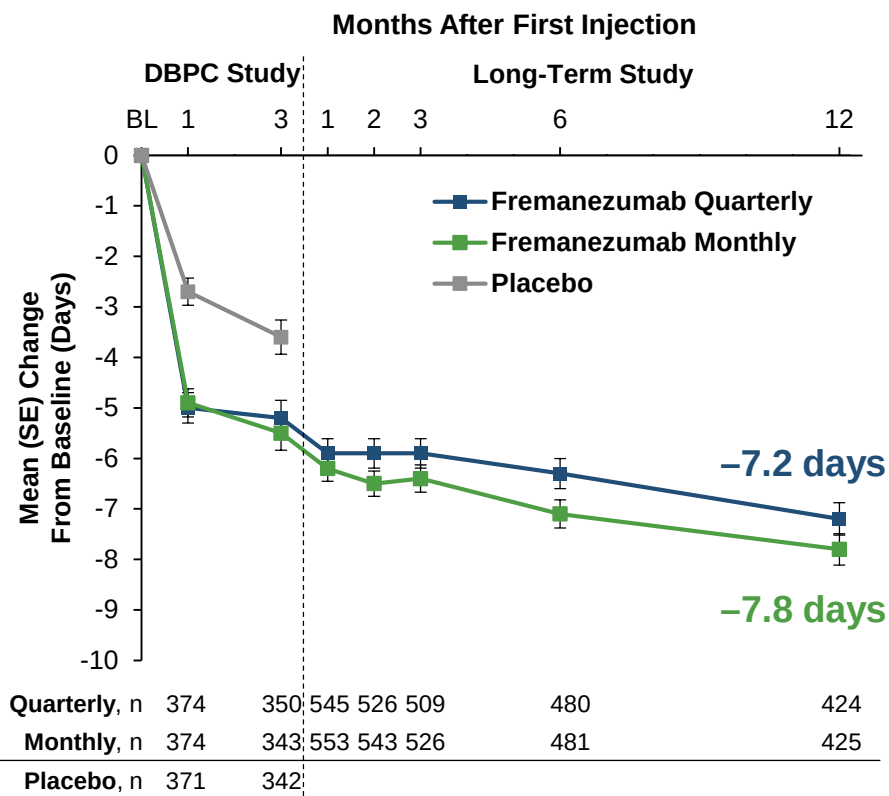


Fremanezumab quarterly or monthly resulted in sustained reductions from baseline in the monthly average number of headache days of at least moderate severity

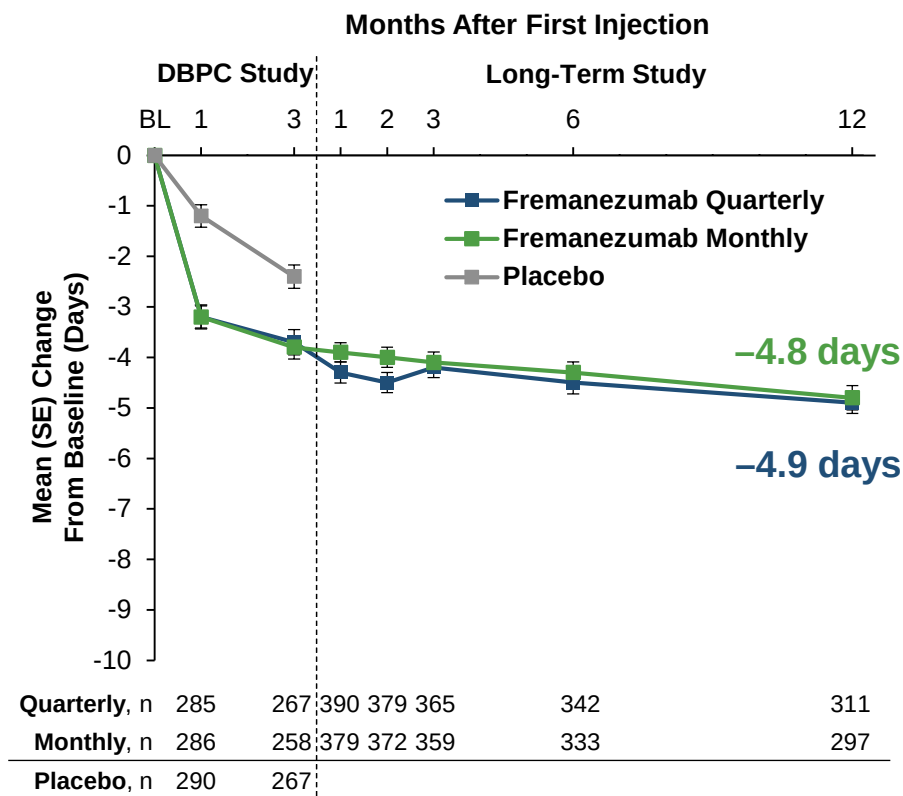
^aIn the FAS population, defined as randomized subjects who received ≥ 1 dose of study drug and had ≥ 1 post-baseline efficacy assessment.
BL, baseline; CM, chronic migraine; DBPC, double-blind, placebo-controlled; EM, episodic migraine; FAS, full analysis set; SE, standard error.

Long-Term Change in Monthly Average Number of Headache Days of Any Severity in Patients With CM and EM^a

Chronic Migraine



Episodic Migraine



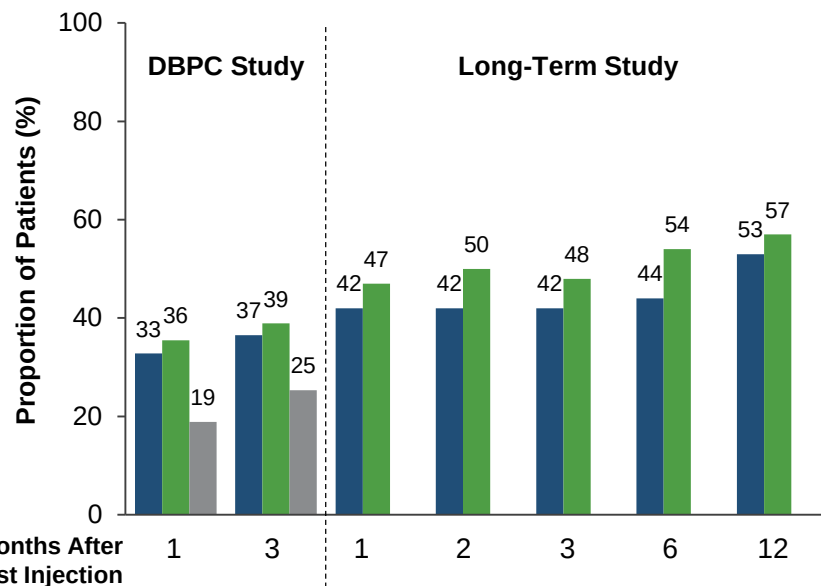
Fremanezumab quarterly or monthly resulted in sustained reductions from baseline in the monthly average number of headache days of any severity

^aIn the FAS population, defined as randomized subjects who received ≥1 dose of study drug and had ≥1 post-baseline efficacy assessment.
BL, baseline; CM, chronic migraine; DBPC, double-blind, placebo-controlled; EM, episodic migraine; FAS, full analysis set; SE, standard error.

Long-Term $\geq 50\%$ Responder Rates for Monthly Number of Migraine Days in Patients With CM and EM^a

Chronic Migraine

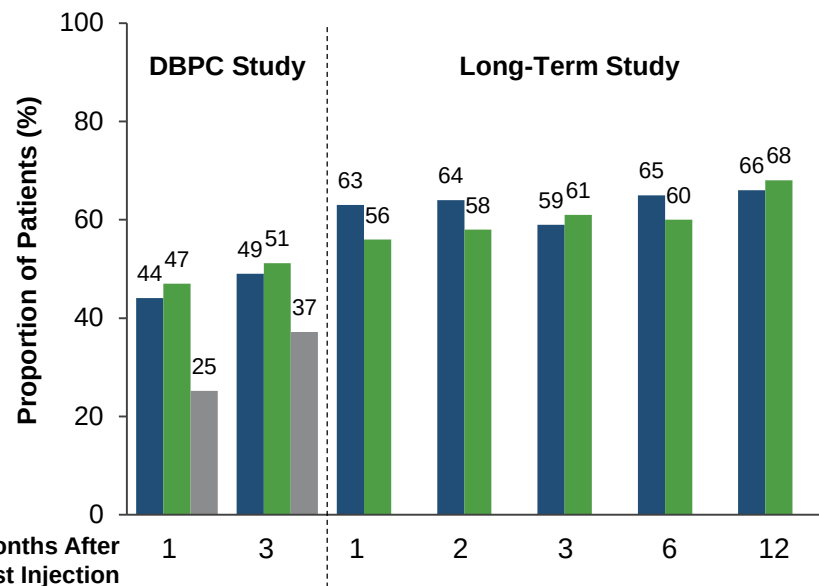
■ Fremanezumab Quarterly ■ Fremanezumab Monthly ■ Placebo



Months After First Injection	1	3	1	2	3	6	12
Quarterly, n	375	350	549	532	515	489	434
Monthly, n	375	346	554	547	531	484	435
Placebo, n	371	343					

Episodic Migraine

■ Fremanezumab Quarterly ■ Fremanezumab Monthly ■ Placebo



Months After First Injection	1	3	1	2	3	6	12
Quarterly, n	288	269	391	379	371	347	313
Monthly, n	287	263	382	372	361	338	299
Placebo, n	290	268					

The proportion of patients with a $\geq 50\%$ reduction in monthly number of migraine days was sustained through Month 12

^aIn the FAS population, defined as randomized subjects who received ≥ 1 dose of study drug and had ≥ 1 post-baseline efficacy assessment. BL, baseline; CM, chronic migraine; DBPC, double-blind, placebo-controlled; EM, episodic migraine; FAS, full analysis set; SE, standard error.

Adverse Events in the Safety Population^a

	CM		EM	
	Fremanezumab		Fremanezumab	
	Quarterly (n=551)	Monthly (n=559)	Quarterly (n=394)	Monthly (n=386)
Patients with AEs, n (%)				
At least one AE	461 (84)	498 (89)	330 (84)	323 (84)
At least one treatment-related AE	299 (54)	328 (59)	213 (54)	223 (58)
At least one serious AE	38 (7)	35 (6)	21 (5)	21 (5)
Any AE leading to study discontinuation	20 (4)	18 (3)	20 (5)	18 (5)
Injection-site reactions (occurring in >6% of patients in any treatment group), n (%)^b				
Injection-site induration	165 (30)	196 (35)	113 (29)	145 (38)
Injection-site pain	157 (29)	182 (33)	118 (30)	123 (32)
Injection-site erythema	138 (25)	171 (31)	85 (22)	103 (27)
Injection-site hemorrhage	42 (8)	44 (8)	17 (4)	28 (7)
Injection-site pruritus	26 (5)	39 (7)	15 (4)	35 (9)
Other common AEs (occurring in >6% of patients in any treatment group), n (%)				
Upper respiratory tract infection	77 (14)	72 (13)	59 (15)	45 (12)
Nasopharyngitis	64 (12)	61 (11)	41 (10)	51 (13)
Sinusitis	40 (7)	39 (7)	19 (5)	18 (5)
Urinary tract infection	39 (7)	28 (5)	22 (6)	24 (6)

- Similar proportions of patients in each fremanezumab treatment arm reported at least one AE
- Injection-site reactions were the most commonly reported AEs, with similar incidence rates between groups
- Serious AEs and AEs leading to discontinuation were infrequent, with similar incidences across treatment groups

^aRandomized subjects who received ≥1 dose of study drug during the study. ^bLocal injection sites were systematically assessed for erythema, induration, ecchymosis, and pain both immediately and 1 hour after dosing.

AE, adverse event; CM, chronic migraine; EM, episodic migraine.

Conclusions

- In patients with CM and EM, long-term treatment (up to 15 months) with fremanezumab quarterly or monthly:
 - Demonstrated an acceptable safety profile
 - Was well tolerated
 - Resulted in sustained reductions from baseline in the monthly number of migraine days, headache days of at least moderate severity, and headache days of any severity
- A clinically significant response to treatment, as measured by a $\geq 50\%$ reduction from baseline in the mean number of monthly migraine days, was achieved as early as Month 1 and was sustained throughout the 12-month treatment period
- These findings support the data from placebo-controlled studies and demonstrate that the efficacy and safety of fremanezumab are maintained over long-term treatment

Acknowledgments

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